

User Guide

For QC Engineers — day-to-day use

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Castillo Planset QC — User Guide for QC Engineers

About this guide

This guide walks the **QC Engineer** through the day-to-day use of the Castillo Planset QC application: signing in, submitting a solar PV planset for automated review, monitoring the analysis, interpreting the findings, managing projects and re-run history, and exporting the results. It is task-oriented — read it top to bottom the first time, then use it as a reference.

The application is an AI-assisted quality-control tool. It renders a planset PDF, runs a combination of per-page and multi-page vision checks, deterministic NEC electrical calculations, and rule-based sheet/title-block/keyword checks, then presents the results as a list of **findings** you triage. **The tool assists your judgment; it does not replace it.** Every automated finding is yours to confirm, correct, or override.

The current in-app version label is **v0.3.0** (shown in the sidebar under the Castillo Engineering brand).

1. Signing in

The application is protected by **Microsoft Entra** single-tenant organization login (your castillope.com account). Access is handled by the Azure "EasyAuth" layer in front of the app.

1. Open the application URL in your browser.
2. If you are not yet authenticated, the top-right corner shows a **Sign in** button. Click it. You are redirected to `/.auth/login/aad` and then to the Microsoft organization sign-in page.
3. Sign in with your castillope.com credentials (and complete MFA if prompted).
4. You are returned to the app. The top-right corner now shows your **account chip** — a circular avatar with your initials plus your display name.

Confirming and managing your identity

- Click the account chip to open the profile menu. It shows your name and email, and a **Sign out** link (which routes through `/.auth/logout`).
- Once signed in, the upload form shows a green "**✓ Signed in as \<your name\>**" confirmation. Runs you start are automatically **attributed** to this Entra identity.
- The **"Your name (QC engineer)"** picker in the upload form defaults to your signed-in name. You may pick a different name from the list, or choose **"+ Add new engineer..."** to add one — but any manual override is **session-only**: it is not persisted, and the next page load reverts to your authenticated identity. The picker list itself (for convenience) is remembered in your browser's local storage.

If you ever see the **Sign in** button reappear, your session expired — sign in again before starting work so your runs are correctly attributed.

2. The main screen layout

The screen has three persistent regions plus overlays.

Region	Location	Contents
Sidebar	Left	The Castillo Engineering brand + version, the upload / analysis form , an All Projects button, and a Recent list of the latest run per lineage. Collapsible with the < / > toggle; on mobile, opened via the ≡ button.
Main content	Center/right	Either the Projects dashboard (a grid of project cards) or a single-run report view , depending on what you have open.
Account chip	Top-right	Your signed-in identity and the sign-out control.
Activity panel	Floating	The shared queue/activity feed of every analysis in progress across all engineers (see §4).
Toasts	Floating	Short-lived completion / error notifications.

The default landing view is the **Projects dashboard**. Opening a run switches the center pane to that run's report; the < **All Projects** button (top-left of the report header) or the sidebar **All Projects** button returns you to the dashboard.

Every run has a **shareable deep link**. In a run's header, click the small **run xxxxxxxx** code to copy a URL (`.../?run=<id>`) to your clipboard; anyone with access who opens it lands directly on that run.

3. Starting an analysis

All inputs live in the sidebar **upload form**. Only a PDF is strictly required; everything else improves the quality and grouping of the results.

Fill the form top to bottom:

5. **Your name (QC engineer)**. Defaults to your signed-in identity. Leave as-is unless you are logging the run for someone else.
6. **Project**. Type a project name, or pick an existing one from the auto-complete list. **Reusing the exact same project name groups the new run under that project** in the dashboard and sidebar. Use this deliberately so all runs for one site stay together.
7. **Run name (optional)**. A label for this specific run (for example, "IFC review — round 2"). If left blank, the run is labeled with the PDF's file name.
8. **The planset PDF**. Drag a PDF onto the **"Drop planset PDF here or click to browse"** zone, or click it to open a file picker. Only PDF files are accepted. The selected file name appears with a clear (x) button if you need to swap it.
9. **Project Details (optional but recommended)**. Click **"▼ Fill Project Details"** to open the details template in the main pane. See §3.1.
10. **Deep mode**. A checkbox, **on by default**.
 - **Deep mode on (hybrid)**: heavy-reasoning checks (single-line diagrams, DC line diagram, three-line diagram, cross-sheet consistency, system-information table) use the full AI model; all other checks use the faster "mini" model. This is the recommended setting for a real QC pass.
 - **Deep mode off (mini only)**: every check uses the mini model — faster and cheaper, lower reasoning depth. Use for quick scans or drafts.
11. **Design stage**. A dropdown that gates which rules apply:
 - **Auto-detect from title block** (default) — the tool infers the stage.
 - **30%, 60% / IFP, 90%, IFC, As-Built** — pin the stage explicitly.

Rules that require **later-stage** sheets are **deferred** (reported as not-applicable-at-this-stage) when you pick an earlier stage. The gating is **lenient**: if a later-stage sheet actually appears in the PDF, its rules still run.

12. **Supporting documents (optional but valuable)**. Drag files onto the **"Supporting documents"** box or use its picker. Accepted types: PDF, Excel (**.xlsx**, **.xls**), CSV, text, email (**.eml**, **.msg**), and images (**.png**, **.jpg**, **.jpeg**). Typical documents: **CESIR, PVSyst report, ampacity/impact studies, equipment datasheets, interconnection agreements**. As each file is read, it appears in the list with a **document-type badge**. Two things happen with these:
 - Findings can be **cross-checked and grounded** against them (you will see a **Source** citation on such findings — file name, page, and a verbatim excerpt).

- Specs the tool extracts (module, inverter, transformer, temperatures, etc.) are **auto-merged into Project Details to fill gaps**, so electrical calculation rules (stringing, fuse sizing, transformer sizing, etc.) can run instead of deferring for lack of data. **Any value you typed yourself always wins over an auto-extracted one.**

13. Click **Analyze PDF**.

3.1 Filling Project Details (the due-diligence template)

The Project Details panel is a structured form covering Project Info, Owner / EPC / Engineering, PV Module, System Configuration, Inverter, Racking, Transformer, Utility / POI, MV / Protection, Design Temperatures, and Special Notes. The tool **compares these declared values against what it reads on the planset and flags mismatches.**

You have three ways to populate it:

- **Type values directly** into any field.
- **Auto-fill from documents:** drop emails, interconnection agreements, CESIR, impact studies, equipment submittals, or any project docs onto the **upload zone at the top of the panel**. The AI reads them and fills empty fields (it does not overwrite values you already entered). Supported: PDF, images, Word, Excel, text, email.
- **Download the Excel template ("↓ Download Excel Template" / "Download Due Diligence Template")** to collect values offline before a review, then transcribe them in.

Use **Clear All** to reset the form, or **Done** to close the panel. When any detail values are set, the **"Fill Project Details"** button shows a small dot indicator.

3.2 What happens when you submit

Submitting does **not** freeze your screen. The run is placed on a **shared, bounded job queue** and processed in the background. The form clears itself, a **"Queued"** toast appears, and you are free to open other runs, queue additional plansets, or return to the dashboard. Progress is tracked entirely through the Activity panel described next.

4. Watching progress: the Activity panel, queue, and notifications

The floating **Activity** panel is the shared view of **every** analysis in progress across all engineers — not just yours. It refreshes automatically every few seconds.

Each row shows a run's name, who started it, and its state:

- **Running** — a live progress bar and percentage, with the current step.

- **Queued** — its position in line (for example, "Queued #2"), because only a bounded number of analyses run at once.
- **Complete** — click the row to open the finished report.
- **Failed** — with a short error detail.

The panel footer summarizes throughput: **"Running N/\<concurrency\> · X queued."** The concurrency limit is a server setting (**ANALYZE_CONCURRENCY**); when it is reached, further runs wait in the queue and start automatically as slots free up.

Completion notifications

For runs **you** started (or that are attributed to your account), the app alerts you when they finish, even if you have navigated away:

- A **toast** appears in the corner with a **View** button that jumps straight to the completed report.
- If the browser tab is in the background, its **title flashes** ("✓ \<run\> done"), reverting when you return to the tab.
- **Desktop alerts (optional):** click " **Desktop alerts**" in the Activity footer and grant your browser's notification permission. You will then get OS-level notifications on completion or failure. The footer shows " **Alerts on**" once enabled.

You are never notified twice for the same run, and reloading the page will not replay alerts for runs that finished while you were away.

5. Reading a report

Open a run (from the dashboard, the Recent list, or an Activity row) to see its report.

5.1 The run header

The header identifies the run and offers its top-level actions:

- **Project name**, plus a **design-stage badge**, a **version tag** (**v2**, **v3**, ...) if it is a re-run, and a **"superseded"** tag if a newer version exists.
- A metadata line: run label, page count, timestamp, attributed engineer, and the copy-to-clipboard **run ID**.
- An **AI usage bar**: number of AI calls, tokens, an approximate cost, total analysis duration, a **Deep / Mini** pill, and a **Stage** pill. An expandable **"Slowest AI calls"** breakdown is available.
- An **Evidence** bar listing the supporting-document types that informed this run.
- Action buttons:  **Rename**,  **Re-analyze** (§7), **Open PDF**, **Export Excel** (§8), **Delete**.

Directly beneath the header, a **"How was this run?"** feedback banner lets you rate the run — 

Saved time,  **About even**, or  **Cost time** — with an optional one-line note. This feedback tunes

the tool over time; rate it when you finish a review. You can dismiss it for the current run, and edit a rating you already gave.

5.2 Score cards

A row of summary cards sits above the findings:

Card	Meaning
Total Checks	All findings emitted.
Actual Checks	Findings actually evaluated (Deferred excluded).
Pass / Fail / Review	Counts by status — click to filter the list.
Deferred	Checks that could not be run at this stage or with the available evidence (shown only when > 0).
Pages	PDF page count.
Complete	Pass rate among <i>evaluable</i> checks (Deferred excluded).
Issues Only	A toggle that filters the list to Fail + Needs Review.

5.3 The four statuses — and what to do with each

Every finding carries a status. The tool sets an initial **auto-status**; your triage sets the **QC status**. Color convention throughout the app:

Status	Color	Meaning	What you should do
Pass	Green	The check was evaluated and the planset satisfies it.	Spot-check as your process requires. No action needed to close.
Fail	Red	The check was evaluated and the planset violates it.	Investigate and confirm. If it is a genuine defect, it becomes a review comment to the designer. If the tool is wrong, correct the status (or Accept , with a note).
Needs Review	Amber	The tool is uncertain — ambiguous drawing, low confidence, or a judgment call it cannot make alone.	This is where your attention is most valuable. Open the finding, look at the cited page/evidence, and

			resolve it to Pass , Fail , or Accept . Do not leave Needs Review items unresolved in a completed review.
Deferred	Purple	The check was not run — either not applicable at the selected design stage, or it needed evidence (a spec, a datasheet, a later-stage sheet) that was not provided.	Not a defect. If a Deferred check <i>should</i> have run, supply the missing supporting document or Project Detail and re-analyze , or pin the correct design stage.

There is a fifth, QC-applied state — **Overridden / Accepted by QC Engineer** (shown as **Accepted**) — which you set when you have reviewed a Fail or Needs Review item and are deliberately clearing it, ideally with a QC note explaining why.

5.4 Severity, confidence, evidence, and page references

Each finding also shows:

- **Severity** — **HIGH**, **MED**, or **LOW**. Use it to prioritize which failures to address first.
- **Confidence** — the tool's self-assessed certainty, shown as a percentage. Low confidence is a cue to look harder; you can **sort by confidence (low → high)** to surface the shakiest findings first.
- **Evidence / AI Finding** — the tool's explanation of what it saw and why it reached its conclusion. Click to expand long text.
- **Source** — when a finding is grounded in a supporting document, the file name, page, and a verbatim excerpt are cited.
- **Page reference** — a **p.N** link that opens the PDF at that page. Findings that span multiple sheets (cross-sheet consistency) show several sheet/page chips and, in the detail modal, a **side-by-side comparison** of the conflicting values.

5.5 Sheet categories

The left **category navigator** organizes findings by planset sheet type — for example, Drawing Index, Title Block, Cover Sheet, System Information Table, General Notes, Site Plan, AC/DC/Three-Line Diagrams, Relay and Inverter Settings, Feeder Plans, Grounding, PVSyst Analysis Summary, Cross-Sheet Consistency, and more. Each category shows a health icon, a pass-progress bar, and per-status counts (**F / R / P / D**). Click a category to filter to it; click one of its status counts to drill directly into, say, that category's Fails.

5.6 Working through the findings list

Above the list, the **toolbar** provides:

- **Status filters** — All / Pass / Fail / Review / Deferred / Accepted (with live counts).
- **Group similar** — collapses multiple instances of the same rule (for example, the same issue on rows 11, 12, 14) into one group header you can expand.
- **Sort** — Default, Severity, Status, Category, Page, or Confidence.
- **? (keyboard help)** and **+ Add Issue (§6)**.

On each **finding card** you can, without opening anything:

- Set the status with the **quick-status buttons** — ✓ (Pass), ✗ (Fail), ? (Needs Review), **OK** (Accept).
- Click the **eye** to open the full-resolution preview modal.
- Click the **pencil** to add or edit a **QC note** (override comment).
- Click the **flag** to **report a problem with the rule itself** (not the QC status) — tagging it as wrong status, wrong page, wrong location, wrong source citation, wrong reason, wrong category, "rule shouldn't exist", or other, with an optional comment. This is separate from triage and feeds tool improvement.

Click the card body to expand its evidence; click the preview thumbnail (or the eye) to open the **detail modal**, which shows the highlighted crop or full page (toggle **fit / actual size** zoom), the evidence and description, and buttons to **Open PDF Page N** and the **Full PDF**.

5.7 Keyboard triage

For fast review, press **?** to see the shortcuts:

Key	Action
j / ↓	Next finding
k / ↑	Previous finding
p	Mark Pass
f	Mark Fail
r	Mark Needs Review
o	Mark Accepted (override)
Enter	Open detail modal
Esc	Close modal / clear focus

Click a card or press **j** to begin; the status keys act on the highlighted (focused) finding.

6. Adding your own findings

The tool will not catch everything. To log a manual observation, click **+ Add Issue** in the toolbar and fill in the category, title, evidence/notes, page, and severity. Manual findings are recorded at full confidence and appear in the list (and the Excel export) alongside the automated ones. They never merge into automated groups.

7. Projects, version history, and re-analyzing

7.1 How work is organized

The dashboard and sidebar group everything as **Project** → **Design Stage** → **Lineage** → **Versions**:

- A **Project** collects all runs that share a project name (or project ID).
- Within a project, runs are grouped by **design stage**.
- A **Lineage** is a chain of re-runs of the same planset (they share a common root).
- A lineage holds one or more **versions**, newest marked **current**.

On the dashboard, each **project card** collapses to a compact summary (stages with **Pass / Fail / Review** pill counts) and expands to show every run, with per-run **Rename**, **Re-analyze**, **version history**, and **Delete** controls. Use **Expand all / Collapse all** and the **search box** (matches project, run, file, and engineer) to navigate. The sidebar's **Recent** list is a quick-switcher to the latest version of each lineage.

7.2 Re-analyzing (creating a new version)

When a designer returns a revised planset, or you want a fresh pass, click **↺ Re-analyze** (in the run header or on the dashboard card). This is **non-destructive**:

- It **saves a new version** and **keeps the prior version in history** (the old one is tagged **superseded**).
- It **carries forward** the original run's Project Details, supporting documents, and design stage, so the re-run has the same signals as the first — not fewer.
- It runs on the same background queue; you are notified on completion.

Expand a lineage's **"N versions"** control to see and open any version (v1, v2, ...), with the latest marked **current**.

7.3 Comparing versions and stages

To answer "did the designer fix the comments?" — or to compare stages — use compare mode:

14. In a run's report, use the **"Compare with:"** picker to choose another run (any run, including a different stage or a different version of the same planset).

15. The list switches to a **diff view** that matches findings by rule and buckets them:

- **Fixed** — was a problem (Fail / Needs Review), now resolved.
- **New** — newly a problem in the current run.
- **Drift** — status changed in some other way.
- **Same** — unchanged.
- **Removed** — a prior problem rule that no longer appears.

16. Use ⇌ to swap before/after, and ✕ to exit compare mode. The comparison is reflected in the URL, so a before/after view can be shared as a link. If the two runs are of different files, the banner warns you that the diff may be misleading.

8. Exporting results to Excel

From a run's header, click **Export Excel** to download a formatted **.xlsx** workbook (named `\<project\>_\<runid\>_qc.xlsx`). It contains:

- **Summary sheet** — project name, original file, created date; top-line metrics (PDF pages, indexed sheets, actual sheets, and counts for Pass / Fail / Needs Review / Deferred / Overridden); and a per-category table (Total, Pass, Fail, Needs Review, Deferred, Overridden).
- **Issues sheet** — one row per finding with **Category, Checklist Item, Status, Auto Status, Severity, Page, Confidence, Evidence, Description, Override Comment**, and an embedded **Snippet** image where available. Status cells are color-coded to match the in-app convention (green / red / amber / purple / blue).

The exported statuses reflect **your** triage decisions (the QC status), with the tool's original **auto-status** preserved in its own column for traceability — so the workbook is a defensible record of both what the tool found and what you concluded.

Separately, the **Download Due Diligence Template** button (on the welcome screen and in the Project Details panel) exports the blank intake spreadsheet for collecting project inputs before a review.

9. Recommended day-to-day workflow

17. **Sign in** and confirm the green "Signed in as" badge.
18. **Set the Project name** (reuse it for the same site) and a descriptive **Run name**.
19. **Attach supporting documents** (CESIR, PVSyst, datasheets) and **fill Project Details** — this is what lets electrical and cross-check rules run instead of deferring.
20. Pick the correct **design stage** (or leave on auto-detect) and keep **Deep mode on** for a real QC pass.

21. **Analyze**, then keep working while the queue runs; you will be alerted on completion.
22. Triage with the keyboard: **clear every Fail and Needs Review** — confirm real defects, correct or **Accept** false positives (with a QC note), and resolve uncertainties against the cited page and evidence.
23. Address **Deferred** items only if a check should have run — supply the missing input and **re-analyze** (which preserves history).
24. **Rate the run** and **Export Excel** for the record; use **Compare** when the revised planset comes back to verify the comments were addressed.